- *Acceptance stage*—The publication of a new release candidate to the artifact repository triggers this stage, which consists of deploying the application to production-like environments and running additional tests to increase the confidence about its releasability. The tests that run in the acceptance stage are usually slow, but we should strive to keep the whole deployment pipeline execution to under one hour. Examples of tests included in this stage are functional acceptance tests and non-functional acceptance tests, such as performance tests, security tests, and compliance tests. If necessary, this stage can also include manual tasks like exploratory and usability tests. At the end of this stage, the release candidate is ready to be deployed to production at any time. If we are still not confident about it, this stage is missing some tests.
- *Production stage*—After a release candidate has gone through the commit and acceptance stages, we are confident enough to deploy it to production. This stage is triggered manually or automatically, depending on whether the organization has decided to adopt a continuous deployment practice. The new release candidate is deployed to a production environment using the same deployment scripts employed (and tested) in the acceptance stage. Optionally, some final automated tests can be run to verify that the deployment was successful.

This section will guide you through bootstrapping a deployment pipeline for Catalog Service and defining the first steps in the commit stage. Then I'll show you how to automate those steps using GitHub Actions.

3.5.1 Understanding the commit stage of the deployment pipeline